Inconvenient Intermissions

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November 9, 2026

Abstract

One-page summary: the problem, your modeling approach, key results, and main recommendation. Write this last.

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1 Introduction

1.1 Background

Seating 1,329, Bailey Hall is Cornell's largest auditorium(source) With such large capacity comes a greater risk for crowd collapses and crushes. While New York state, Ithaca, and Cornell(source) enforce regulations and evacuation protocols to mitigate these risks, the presence of an emergency like a fire or invader can lead to crowd dynamics which are difficult to predict and control. In order to best ensure safety of Bailey patrons, we must understand how these dynamics arise at an individual and collective level.

1.2 Problem Restatement

We are asked to use mathematical modeling to answer five questions about egress from Bailey Hall:

1. How long does it take a concert audience to leave Bailey Hall?
2. How long does it take the same audience to leave in an emergency when guided by Crowd Managers, and how does that time change if one exit is unexpectedly obstructed?
3. How does this compare with evacuating the same number of students from a lecture, where no Crowd Managers are present but students know the building and its emergency exits?
4. Could egress be made faster, safely, with removable barriers in the corridor that route people from the mezzanine exits to specific exterior doors, or with directional LED strips that guide parts of the audience to specific exits?
5. Are such measures cost-effective compared with staffing more Crowd Managers?

1.3 Literature Review

Fortunate for us, there is already a large existing literature on Crowd analysis. Various authors have considered both discrete (source) and continuous (source) approaches to modeling the dynamics of large crowds. Of particular interest to us where papers on Floor Field Cellular Automata (FFCA) models (source), and (continuous source).

We found the FFCA framework to be useful because it allowed us to model the evolution of a threat, collective interests and individual decision making all at once. In this framework, a space (floor) is modeled as a grid. Static and dynamic fields over this grid carry information about things like proximity to an exit, or fire, traversability and crowd trends. Individuals are modeled as cells, who occupy a space on the grid and make a decision about where to move at every timestep based on their own characteristics and the various fields in use.

1.4 Our Approach

We developed two complementary models. One based on the FFCA framework described above, and another dynamical system which models how crowd density and "threat" (from a fire or invader) develop throughout the duration of an emergency.

2 Assumptions and Justifications

2.1 Patrons

Assumption 1. Our first assumption is that the average person behaves approximately optimally when exiting Bailey under normal conditions. We believe this is justified as most people have been in enough

buildings and have enough common courtesy to work together in a timely manner. Here we are only concerned with optimality of homogeneous behaviors of crowds.

Assumption 2. Our second assumption is that in the event of an emergency people's behavior is consistent with their behavior in the absence of an emergency. In our context this means that the exit and crowd weights k_S and k_D do not change when we introduce the fire weight k_F or ushers; an emergency changes the information people act on, not how they act on it. (There's something on the wiki to justify this)

Assumption 3. Our third assumption is that every Bailey patron is able bodied. The reasoning behind this assumption is that accounting for wheelchair accessibility would add complexity to our model (can probably justify this better)

Assumption 4. Our fourth assumption is that every Bailey patron is seated on the first floor, and that every one of its 1010 seats is occupied when the emergency begins. A full house is the worst case for an evacuation.

Assumption 5. Our fifth assumption is that each patron occupies one 0.432 m by 0.432 m cell, and that no two patrons can share a cell.

Assumption 6. Our sixth assumption is that every Bailey patron moves at the same speed.

Assumption 7. Every patron walks at 1.2 m/s (about 2.7 mph), so one time step, the time to walk one cell orthogonally, is $\Delta t = 0.432\,\text{m}/1.2\,\text{m/s} = 0.36\,\text{s}$. A diagonal step also takes one time step.

Assumption 8. Every patron knows the layout of the hall and the shortest route from their position to the nearest exit. Patrons at a concert or lecture entered through the same doors and corridors, and the exit doors are visible from the seating.

Assumption 9. Every patron starts moving at the start of the emergency, with no delay to notice the alarm or gather belongings. This makes our evacuation times lower bounds.

Assumption 10. Patrons differ in their decision weights. Their exit and crowd weights (k_S, k_D) are normally distributed around the optimum for a homogeneous crowd, with covariance $c\,H^{-1}$ and $c = 1$, so that each patron's behavior, if adopted by a whole crowd, would cost on average one extra time step. Their fire weights k_F are lognormally distributed with spread $\sigma_F = 0.5$, independently of (k_S, k_D).

2.2 Movement

Assumption 11. All patrons decide their moves at the same time, based on the state of the hall at the start of the time step. A patron cannot move into a cell that another patron is leaving in the same step.

Assumption 12. When several patrons try to enter the same cell, with probability $\mu = 0.3$ none of them succeeds, and otherwise one of them does, with probability proportional to how strongly each preferred that cell. The trail left by patrons diffuses and decays at rates $\alpha = \delta = 0.3$ per time step.

Assumption 13. Each exit door admits at most one patron per time step, with probability 0.6, about 1.3 people per meter of door width per second. Patrons are safe once they pass through an exit door; congestion in the corridors and lobby beyond is not modeled.

2.3 Geometry

Assumption 14. The geometry of the first floor is our parametric reconstruction of the seating chart. Row spacing, the width of the center section and the positions and widths of the six access points are measured from the chart; the remaining dimensions are estimates. The positions of the nine exit doors, each 1.3 m wide and evenly spaced along the back wall, are placeholders in the absence of evacuation plans.

2.4 Fire

Assumption 15. Fire spreads only between adjacent cells, following rules adapted from Minecraft, and its spread does not depend on the patrons.

Assumption 16. Every non-wall cell (seats, legroom, aisles, access points and exits) is equally flammable, with the properties of wool in Minecraft. Walls are stone and never burn.

Assumption 17. Each fire starts in a single cell, chosen uniformly at random among the non-wall, non-exit, unoccupied cells, at the moment the evacuation begins.

Assumption 18. The fire represents untenable conditions (flames and smoke together), which spread through the hall in a median of 6 min.

Assumption 19. Patrons sense the fire within 3.5 m and react to it in proportion to how many steps away from it a move takes them. Beyond that distance the fire does not affect their decisions.

Assumption 20. A patron who spends 5 s in total in burning cells is incapacitated and remains in place as an obstacle to others. Patrons still inside after 12 min are counted as trapped. Both count as casualties.

2.5 Ushers

Assumption 21. The 6 ushers stand against the outer wall of the hall from the start of the emergency, do not occupy a cell or block patrons, and are not affected by the fire.

Assumption 22. An usher influences patrons within $R = 10$ cells (about 4.3 m) of walking distance, and a patron in range of several ushers follows the nearest. A patron instructed by an usher remembers the instruction for the rest of the evacuation.

Assumption 23. Each usher directs patrons to one assigned exit door, which does not change during the evacuation, even if the fire reaches it.

Assumption 24. Patrons differ in how closely they follow ushers. Each patron's compliance is drawn independently from a Beta distribution with mean $\bar{w}$ and concentration 4, independently of their decision weights, and a compliant patron pursues the assigned door with the same determination k_S they would use for the nearest exit.

3 Notation

Symbol	Meaning
SD_{ij}	steps from cell (i, j) to the nearest exit
$SD_{ij}^{(E)}$	steps from cell (i, j) to exit door E
ST_{ij}	traversability of cell (i, j) (1 or 0)
D_{ij}	dynamic (foot traffic) field
F_{ij}	fire proximity field
n_{ij}	occupancy of cell (i, j) (1 or 0)
k_S, k_D, k_F	a patron's weights on SD, D and F
$x^* = (k_S^*, k_D^*)$	optimal exit and crowd weights
H	Hessian of mean exit time at x^*
c	spread of the population around x^*
m, σ_F	median and spread of k_F
μ	friction: probability no one wins a contested cell
α, δ	diffusion and decay of D
A_c	age of a burning cell
s, q	fire spread scale and update probability
ℓ	fire sensing range
$w_a, \bar{w}$	compliance of patron a; mean compliance
R	usher range
Δt	time step, 0.36 s
T	simulation horizon, 2000 steps

4 Model

4.1 FFCA

4.1.1 Baseline Model

In our FFCA model, we start by rendering the first floor of Bailey hall as a 60 by 78 grid of .432 meter by .432 meter cells. We chose this cell size based on (x) which suggest that the average human occupies an area of .43 meters by .43 meters (source to justify). We then assign one of six types to each grid cell: wall, aisle, legroom, seat, access point, or exit. Aisle, legroom, access point and exit cells are traversable; walls and seats are in-traversable, although a seated patron may step out of their seat. Based on these types we create a distance static field SD and traversability static field ST, defined by

$$SD_{ij} = \text{number of steps on the shortest path to an exit via traversable cells}$$

$$ST_{ij} = \begin{cases} 1 \text{ if cell (i,j) is traversable} \\ 0 \text{ otherwise} \end{cases}$$

The idea here is that SD incorporates information about proximity to the nearest exit, and that ST incorporates information about which grid cells can be walked through. To actually calculate SD we used a breadth first search from the exit cells, in which a step to any of the eight neighboring cells (including diagonals) counts as one step and a diagonal step may not cut the corner of a wall. A seat cell is assigned one more than the smallest value of SD among its traversable neighbors. We computed ST from the cell types, which come from our reconstruction of the seating chart.

Next, we define a dynamic field D as follows. During a given time step, if an automaton (patron) exits cell (i, j) we increment D_{ij} by one unit. Then, after every patron has attempted to move, a fraction α of each cell's value D_{ij} "dissipates" equally to its traversable up, down, left and right neighbors, and then every value "decays" by a fraction δ, i.e. $D_{ij} \leftarrow (1 - \delta)D_{ij}$. We use $\alpha = \delta = 0.3$. (reference whatever paper this comes from, one of the ones in the doc)

Finally, for a patron a located at grid cell (i, j) the probability of transfer to the cell (i', j'), where (i', j') is one of the eight neighboring cells or (i, j) itself (staying put), during a timestep is given by:

$$p^{(a)}_{i',j'} = N^{(a)}_{i,j} \exp(-k^{(a)}_S SD_{i',j'}) \exp(k^{(a)}_D D_{i',j'}) ST_{i',j'}(1 - n_{i',j'}) \tag{1}$$

where $N^{(a)}_{i,j}$ is a normalization constant chosen so that these probabilities sum to one.

$n_{i',j'}$ is a binary variable which is 1 when the cell (i', j') is occupied and 0 otherwise (a patron's own cell always counts as unoccupied); the idea here is that a cannot decide to move to a space that is currently occupied even in the event that the occupant concurrently decides to vacate the spot. We believe this is justified and aligns with behavior seen for example in traffic jams(source on ghost traffic or something).

$k^{(a)}_S$ and $k^{(a)}_D$ are a's decision weights, and are meant to represent how a values different information given by the fields: k_S is how strongly a heads toward an exit, and k_D is how strongly a follows ($k_D > 0$) or avoids ($k_D < 0$) recent foot traffic. We based this formulation on results from (source).

At each step in the simulation, every patron a simultaneously and randomly chooses which of the nine candidate cells to move to based on Equation (1). While we saw various approaches as to how collisions are handled in the literature (sources), we decided that when several patrons choose the same cell, with probability $\mu = 0.3$ none of them moves, and otherwise one of them moves while the others stay put, chosen at random with probability proportional to each patron's probability of choosing that cell. We chose this rule because it captures some idea of conviction, i.e. someone who is more confident about where to move next is more likely to beat out those who are less confident, as well as friction between people competing for the same space.

Each exit door can admit at most one patron per time step, and only with probability 0.6, which

corresponds to about 1.3 people per meter of door width per second, consistent with measured flows through bottlenecks (source).

Finally, if a person reaches an exit cell, it is removed from the simulation.

4.1.2 Fire Dynamic Field

In this section we extend our baseline model with a fire that starts at a random location and spreads while patrons evacuate. While there is an existing literature on how fire spreads, especially outdoors in wildfires, we found few discrete models suitable for a building interior, so we based our fire on the fire mechanics of everyone's favorite game, Minecraft (Java Edition) [1], restricted to two dimensions. The fire makes cells impassable, repels patrons through a fire proximity field, and incapacitates patrons who remain in it too long.

Fire spread. Each cell c carries an age $A_c \in \{-1, 0, \dots, 15\}$, where $A_c = -1$ means c is not burning, and a flag recording whether its fuel has been used up. A cell is flammable if it is not a wall, not burning, and still has fuel. Let $M(c)$ denote the eight cells surrounding c. At each time step the fire updates with probability q, analogous to Minecraft's random block ticks; when it updates, every cell changes simultaneously:

1. A burning cell c' ignites each flammable neighbor independently with probability

$$p_{\text{ign}}(A_{c'}) = s \cdot \frac{\iota + 40 + 7\kappa}{100\,(A_{c'} + 30)}, \tag{2}$$

   so a flammable cell c catches fire with probability $1 - \prod_{c' \in M(c),\, A_{c'} \geq 0} \big(1 - p_{\text{ign}}(A_{c'})\big)$.

2. A burning cell uses up its fuel with probability $\beta/300$.

3. The age of a burning cell increases by one with probability $1/3$, up to 15.

4. A burning cell goes out if $A_c > 3$ and no neighbor is flammable, or with probability $1/4$ once $A_c = 15$.

A cell whose fire is consumed or burns out becomes ash, which is passable but no longer flammable. These rules follow Minecraft, with two changes forced by working in two dimensions: Minecraft's fire is a block next to separate fuel blocks and can spread upward, whereas ours occupies the fuel cell itself and spreads within its eight-cell neighborhood. The fuel constants are those of wool, ignite odds $\iota = 30$ and burn odds $\beta = 60$, at the normal difficulty level $\kappa = 2$ [1].

Calibration. The spread scale s controls the fire's shape, whether an ignition takes hold and how completely the front burns the hall, while the update probability q controls only its speed: lowering q stretches the fire's timeline by $1/q$ without changing how it spreads. We calibrate them separately. For each candidate s we run the fire alone from 40 random ignition sites with $q = 1$. A fire is established if it eventually reaches at least half the non-wall cells, and its engulfment time is the first update at which it has reached 95% of the area it eventually burns. Below a threshold value of s most ignitions fizzle, so lowering s cannot make the fire slower without making it unreliable. We choose $s = 8$, for which about 90% of ignitions become established fires that burn essentially the whole hall; the rest fizzle, as some real ignitions do. With the median engulfment time n_{eng} of established fires, measured in updates, a target engulfment time T_{eng} requires

$$q = \frac{n_{\text{eng}}\,\Delta t}{T_{\text{eng}}}. \tag{3}$$

We take $T_{\text{eng}} = 6\,\text{min}$, which with $n_{\text{eng}} = 78$ gives $q = 0.078$. For comparison, fire-protection practice models fire growth as a t^2 design fire [2], under which the flame front in upholstered seating advances only millimeters per second. Smoke, not flame, is the main hazard in real assembly fires and fills a hall far faster [2], so our fire represents the spread of untenable conditions, flames and smoke together, rather than flames alone.

Fire proximity field. Let r_{ij} be the number of steps from (i, j) to the nearest burning cell through non-wall cells. The fire proximity field is

$$F_{ij} = \max\big(0,\ \ell - r_{ij}\big), \qquad \ell = 8 \text{ cells } (\approx 3.5\,\text{m}). \tag{4}$$

Within ℓ cells of the fire, each step away from it lowers F by exactly one, the same unit in which each step toward an exit lowers SD, so k_F/k_S is how much a patron values a step away from the fire relative to a step toward an exit.

Movement and casualties. With fire present, the transition probability is

$$p^{(a)}_{i',j'} = N^{(a)}_{i,j} \exp\big(-k^{(a)}_S SD_{i',j'} + k^{(a)}_D D_{i',j'} - k^{(a)}_F F_{i',j'}\big) ST_{i',j'}\,(1 - n_{i',j'})\,\mathbf{1}[(i', j') \text{ not burning}]. \tag{5}$$

Burning cells and casualties block paths, so SD is recomputed whenever either changes; cells from which no exit can be reached are assigned one more than the largest finite distance. A patron standing in a burning cell for 5 s in total becomes a casualty and remains in place as an obstacle; patrons still inside at T are counted as trapped. Each simulation starts with a single ignition cell drawn uniformly from the non-wall, non-exit, unoccupied cells.

Heterogeneous population. Each patron a receives individual weights. By Assumption 1, we draw (k_S, k_D) around the optimum x^* for a homogeneous crowd (Section 5.1.1). Near x^*, mean exit time is approximately $J(x) \approx J^* + \frac{1}{2}(x - x^*)^\top H(x - x^*)$, and we draw

$$\big(k^{(a)}_S, k^{(a)}_D\big) \sim \mathcal{N}\big(x^*,\ c\,H^{-1}\big), \tag{6}$$

redrawing any sample with $k_S \le 0$. Patrons then vary widely where variation is cheap and narrowly where it is costly, and $\mathbb{E}[J(x^{(a)}) - J^*] \approx \frac{1}{2}\operatorname{tr}(H\,cH^{-1}) = c$, so c is the average cost per patron in time steps. Equivalently, Equation (6) is the quadratic approximation of a Boltzmann distribution $p(x) \propto \exp(-(J(x) - J^*)/c)$. We use $c = 1$. The fire weight is lognormal, independent of (k_S, k_D):

$$k^{(a)}_F = m\, e^{\sigma_F z_a}, \qquad z_a \sim \mathcal{N}(0, 1),\ \sigma_F = 0.5. \tag{7}$$

Learning the fire weight. Holding the distribution of (k_S, k_D) fixed (Assumption 2), we learn the median m by minimizing

$$\mathcal{J}(m) = \mathbb{E}\left[C + \lambda\,\frac{\bar{t}}{t_{\text{ref}}}\right], \qquad \lambda = 0.1,\ t_{\text{ref}} = 174 \text{ steps}, \tag{8}$$

where C is the casualty fraction, $\bar{t}$ the survivors' mean exit time, and t_{ref} the optimal mean exit time without fire, so one percent of casualties counts as much as about 17 steps of exit time. We evaluate $m \in \{0, 0.5, 1, 2, 3, 5, 8, 12, 20\}$ with 50 replicates each. Every replicate fixes the ignition site, the population draw, the patrons' random stream and a separate random stream for the fire, all shared across values of m (common random numbers); since the fire does not depend on the patrons, each replicate's fire is identical for every m, and differences between values of m are paired comparisons.

4.1.3 Usher Static Field

Real evacuations are rarely left entirely to the crowd: ushers direct patrons toward particular exits, typically to spread people across doors that would otherwise go unused. We add 6 ushers to the model. Usher u stands at a cell b_u on the outer edge of the hall, next to the wall, and directs patrons toward an assigned exit door E_u. Ushers stand against the wall and do not occupy a cell, so they never block a patron's path.

For an exit door E, let $SD_{ij}^{(E)}$ be the distance from (i, j) to the cells of door E only, computed in the same way as SD (breadth first search through traversable cells, every step to one of the eight neighboring cells counting as one step). Directing a patron to a door also directs them through whichever access point lies on their shortest path to that door, so ushers can redistribute orchestra patrons across the access points as well as across the doors.

An usher can only reach patrons nearby. The influence region U_u of usher u is the set of traversable cells within R steps of b_u, measured through traversable cells so that walls block it. We use $R = 10$ cells (about 4.3 m), roughly the distance at which an usher can make themselves heard and seen in a crowd. Where two regions overlap, the cell belongs to the nearer usher. A patron becomes instructed by usher u the first time they stand in U_u, and follows that instruction for the rest of the evacuation; we write $E(a)$ for the door patron a was directed to.

Patrons do not all comply equally. Each patron a receives a compliance $w_a \in [0, 1]$, drawn once at the start of the simulation from a Beta distribution with mean $\bar{w}$ and concentration 4,

$$w_a \sim \mathrm{Beta}\big(4\bar{w},\ 4(1-\bar{w})\big). \tag{9}$$

A patron with $w_a = 0$ ignores the ushers entirely and a patron with $w_a = 1$ follows them completely. Since there is little data on how closely people follow staff instructions in an emergency, we treat the mean compliance $\bar{w}$ as a scenario parameter and compare several values rather than fixing one.

An instructed patron blends their own goal, the nearest exit, with the usher's:

$$SD_{ij}^{(a)} = (1 - w_a)\, SD_{ij} + w_a\, SD_{ij}^{(E(a))}, \tag{10}$$

and an uninstructed patron has $SD^{(a)} = SD$. The transition probability is Equation (1) with SD replaced by $SD^{(a)}$:

$$p_{i',j'}^{(a)} = N_{i,j}^{(a)} \exp(-k_S^{(a)} SD_{i',j'}^{(a)}) \exp(k_D^{(a)} D_{i',j'}) ST_{i',j'} (1 - n_{i',j'}). \tag{11}$$

Because the term $-k_S\, SD^{(a)}$ keeps its sign, an instructed patron prefers moves that bring them closer to a mix of the nearest exit and the assigned door: a step that brings them one cell closer to the assigned door, but no closer to the nearest exit, lowers $SD^{(a)}$ by w_a and so is favored by a factor $e^{k_S w_a}$. A fully compliant patron heads for the assigned door with the same determination k_S they would otherwise use to head for their nearest exit. Ushers therefore introduce no new behavioral weight: they change the information patrons act on, not how patrons act on it, consistent with Assumption 2.

When a fire is present, burning cells and casualties block $SD^{(E)}$ exactly as they block SD, and $SD^{(E)}$ is recomputed whenever SD is. An usher's assigned door does not change during the evacuation, so a patron directed toward a door that the fire has cut off simply faces a longer path.

The ushers' positions b_u are chosen from the walkable cells along the outer wall of the hall, and each usher's door E_u from the doors nearest to it. Choosing the positions and doors that minimize casualties is a combinatorial problem, which we treat in Section 4.1.4.

4.1.4 Combinatorial Optimization

We choose the positions and doors of the 6 ushers to minimize the fire objective Equation (8), with the population Equation (6), the learned median fire weight $m = 1$, and mean compliance $\bar{w} = 0.5$. Candidate positions are the walkable cells along the outer wall, thinned to at least 4 cells apart, which gives 40 slots; each usher may direct patrons to either of the two doors nearest its slot. The 80^6 possible configurations are far too many to evaluate, so we search heuristically:

1. *Greedy placement.* We add ushers one at a time, each time choosing the slot and door that most reduce $\mathcal{J}$ given the ushers already placed, estimated on 12 fire scenarios with common random numbers.

2. *Local search.* For each usher in turn, we try moving it to one of the two slots on either side along the

wall, or switching its door, and accept a move only if it improves $\mathcal{J}$ by more than one paired standard error. We repeat for up to three passes, stopping when a pass finds no improvement.

3. *Evaluation.* Because the search selects the best of many noisy estimates, its own estimates are optimistic. We therefore re-evaluate the final configuration on 100 fresh scenarios not used in the search, alongside no ushers, the greedy configuration and 10 random configurations of 6 ushers.

4. *Marginal value.* To measure what each additional usher contributes, we evaluate the first k greedy ushers for $k = 0, \ldots, 6$ on the same fresh scenarios.

5 Methodology

5.1 Determining the Coefficients

The model has three kinds of coefficients. The behavioral weights k_S, k_D and k_F describe how patrons decide where to move, and we learn them by simulation. The fire parameters s and q are calibrated against a target fire behavior. The remaining constants are fixed from the literature or by assumption. Table 1 summarizes how each is determined.

5.1.1 Exit and Crowd Weights

By Assumption 1, patrons behave approximately optimally when leaving Bailey under normal conditions. We therefore take (k_S, k_D) to be the values that evacuate a full hall fastest, for a homogeneous crowd in which every patron has the same weights and there is no fire.

Objective. For one simulation with all 1010 seats occupied, let t_a be the time step at which patron a exits. We minimize the expected mean exit time

$$J(k_S, k_D) = \mathbb{E}\left[\frac{1}{1010}\sum_a \min(t_a,\ T+1)\right], \tag{12}$$

where patrons still inside at the horizon $T = 2000$ steps count as exiting at $T + 1$. This is equivalent to maximizing the area under the curve of patrons evacuated against time, so it rewards getting people out early, penalizes patrons who never leave, and is much less noisy than the time at which the last patron exits. The expectation is over the patrons' random choices and the collision and door draws.

Why both weights are free. The transition probabilities Equation (1) depend on (k_S, k_D) in two ways: through their ratio, which sets how patrons trade exit-seeking against crowd-following, and through their overall magnitude, which sets how deterministic patrons are. Constraining the weights, for example to $k_S + k_D = 1$, would fix the magnitude and remove the second effect, which turned out to matter most. We therefore search over both weights without constraints, allowing either sign for k_D.

Coarse search. We evaluate J on the grid $k_S \in \{0, 1, \ldots, 10\}$, $k_D \in \{-8, -7, \ldots, 4\}$, with $R = 20$ independent simulations (replicates) per grid point. Replicate r uses the same random seed at every grid point (common random numbers), so differences between grid points reflect the weights rather than different random draws.

Refinement. Around the best coarse point we evaluate a finer 9×9 grid spanning one coarse step on each side, again with 20 replicates and common random numbers. Because the objective is noisy, the best point on the refined grid is not a reliable estimate of the optimum: it is the minimum of 81 noisy averages, and is biased toward values that happened to be lucky. Instead we fit the quadratic

$$J(k_S, k_D) \approx c_0 + c_1 k_S + c_2 k_D + c_3 k_S^2 + c_4 k_S k_D + c_5 k_D^2 \tag{13}$$

by least squares to all 81 refined averages. Its minimizer $x^* = (k_S^*, k_D^*)$ solves

$$Hx^* = -\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}, \qquad H = \begin{pmatrix} 2c_3 & c_4 \\ c_4 & 2c_5 \end{pmatrix}, \tag{14}$$

and is a minimum because both eigenvalues of the Hessian H are positive. We check that x^* lies inside the refined grid, so the fit interpolates rather than extrapolates.

Uncertainty. To obtain confidence intervals for x^* we bootstrap over replicates: we resample the 20 replicate indices with replacement, using the same resampled indices at every grid point to preserve the common random numbers, refit the quadratic, and record the minimizer. The 2.5th and 97.5th percentiles over 2000 resamples give 95% intervals. The eigenvalues of H also describe the shape of the optimum: a large eigenvalue means J rises quickly in the direction of its eigenvector, and a small one means it is nearly flat. A small eigenvalue produces a long, narrow valley of near-optimal weights, along which the individual weights are poorly determined even though the valley itself is well determined.

Validation. To confirm that the fitted optimum reflects a real effect rather than noise, we compare it directly with nearby alternatives on fresh random seeds with 200 replicates. For instance, to test whether crowd avoidance helps, we compare $k_D = k_D^*$ with $k_D = 0$ at $k_S = k_S^*$, using the same seeds for both and computing the standard error of the paired difference.

5.1.2 Population Spread

The fitted x^* and H define the heterogeneous population Equation (6). Its spread c cannot be learned in the same way: any spread is worse than none for a crowd optimized as a whole, so optimizing c would remove the heterogeneity. We instead fix $c = 1$, so each patron's weights would, if adopted by a whole crowd, cost on average one extra time step.

5.1.3 Fire Weight

The fire weight k_F is learned as the median m of its lognormal distribution Equation (7), with the population of (k_S, k_D) held fixed and the objective Equation (8) averaged over random ignition sites, as described in Section 4.1.2. The spread $\sigma_F = 0.5$ is fixed for the same reason as c.

5.1.4 Fixed Constants

The friction $\mu = 0.3$ and the dynamic field rates $\alpha = \delta = 0.3$ are fixed at values typical of floor field models (sources). Door throughput, at most one patron per step with probability 0.6, is set so that each 1.3 m door passes about 1.3 people per meter per second, within the range measured for bottlenecks [3]. The fire parameters s and q are calibrated as in Section 4.1.2.

Table 1: How each coefficient is determined.

Coefficient	Meaning	How determined
k_S	weight on distance to an exit	learned: optimal homogeneous crowd, no fire
k_D	weight on recent foot traffic	learned: optimal homogeneous crowd, no fire
H	curvature of J at the optimum	fitted quadratic, Equation (13)
c	spread of (k_S, k_D) in the population	assumed: 1 time step
k_F (median m)	weight on fire proximity	learned: minimizes Equation (8)
σ_F	spread of k_F	assumed: 0.5
s	fire spread scale	calibrated: about 90% of ignitions establish
q	fire update probability	calibrated: median engulfment in 6 min
μ	friction	fixed from the literature
α, δ	dynamic field diffusion, decay	fixed from the literature
door throughput	one patron w.p. 0.6 per step	matched to measured bottleneck flow

6 Results

6.1 FFCA

All simulations use the full house of 1010 patrons on the first floor. Times are in time steps of $\Delta t = 0.36\,\mathrm{s}$ (Assumption 7).

6.1.1 Baseline Model

Optimal exit and crowd weights. The coarse search found its best point at $(k_S, k_D) = (2, -3)$, with mean exit time 173.8 ± 0.7 steps. The quadratic fit to the refined grid gives the optimum

$$(k_S^*, k_D^*) = (2.08,\ -2.98), \tag{15}$$

with 95% bootstrap intervals $k_S^* \in [2.07, 2.12]$ and $k_D^* \in [-3.33, -2.82]$ and predicted optimal mean exit time $J^* = 174.3$ steps (63 s). Both Hessian eigenvalues are positive (1.0 and 29.9), so this is a genuine minimum. On 200 fresh paired replicates, crowd avoidance ($k_D = -2.98$) beat $k_D = 0$ at $k_S = 2.08$ by 7.4 ± 0.3 steps (182.5 versus 175.1), confirming the effect.

Interpretation. Optimal patrons are moderately goal-directed and avoid recent foot traffic. With $k_S^* \approx 2.08$, a cell one step closer to an exit is favored by a factor $e^{2.08} \approx 8$. More strongly goal-directed patrons converge on the same aisles, access points and doors and queue there while other routes sit underused; negative k_D spreads patrons onto less-traveled routes. This depends on the doors' limited throughput: without a limit, mean exit time kept decreasing as k_S grew and there was no interior optimum.

Access points. With the access points at their measured positions and widths, the optimal mean exit time is 174 steps, compared with 150 in an earlier grid with evenly spaced placeholder access points, while the optimal weights barely changed. The access points are the main bottleneck for the 525 orchestra patrons, and their placement alone changes the evacuation time by about 16%.

Population. With $c = 1$, the population Equation (6) has $\mathrm{sd}(k_S) \approx 0.20$, $\mathrm{sd}(k_D) \approx 1.0$ and $\mathrm{corr}(k_S, k_D) \approx -0.45$.

6.1.2 Fire Dynamic Field

Calibration. Table 2 shows the fire for several spread scales with $q = 1$.

Table 2: Fire calibration ($q = 1$, 40 random ignitions per s). Coverage and engulfment are medians over established fires; the last column is the q giving a median engulfment time of 180 s.

s	established	coverage	engulfment (updates)	q for 180 s
4	78%	0.91	174	0.348
6	88%	0.99	100	0.200
8	90%	1.00	78	0.156
12	100%	1.00	62	0.124
16	100%	1.00	53	0.106

Optimal fire weight. Table 3 shows the results. Moderate fire avoidance reduces casualties by about a third: raising m from 0 to 1 lowers the casualty fraction from 2.5% to 1.6% (from about 25 to 16 of 1010 patrons). Strong avoidance is counterproductive: casualties rise to about 3.5% for $m \geq 8$, and the share of

trapped patrons more than doubles, as patrons who put distance from the fire ahead of progress toward an exit are cut off. Avoidance also delays survivors, whose mean exit time rises from 176.5 steps at $m = 0$ to 212 at $m = 20$. $\mathcal{J}$ is minimized at $m = 1$, with $m = 0.5$ within two paired standard errors. At $m = 1$ the typical patron weights a step away from the fire about half as much as a step toward an exit.

Table 3: Learning the median fire weight m ($T_{\text{eng}} = 6\,\text{min}$, 50 paired replicates). Exit time is the survivors' mean, in steps.

m	$\mathcal{J}$	casualties	burned	trapped	exit time
0	0.126	2.49%	2.27%	0.22%	176.5
0.5	0.122	1.90%	1.72%	0.18%	178.3
1	0.120	1.61%	1.43%	0.19%	180.5
2	0.127	1.82%	1.50%	0.33%	188.4
3	0.136	2.26%	1.86%	0.40%	196.4
5	0.148	2.91%	2.43%	0.48%	207.3
8	0.155	3.38%	2.82%	0.56%	210.7
12	0.157	3.49%	2.94%	0.55%	211.5
20	0.157	3.53%	2.97%	0.56%	212.0

6.1.3 Usher Static Field

Door loads without ushers. Without ushers, exits are used very unevenly (Table 4). The two outermost doors carry 44% of all patrons and the center door 3%; at 0.6 patrons per step, each outermost door needs at least 370 steps (130 s) to clear its load, so these two doors set the total egress time. The steps by the stage are almost unused, since they lead up into the boxes, the long way round to the busy outer doors.

Table 4: Mean number of patrons using each door and access point, fire scenarios, no ushers (50 replicates). Pairs are left and right.

Door or access point	position x (m)	patrons
outermost doors	±14.7	221, 222
second doors	±11.6	108, 110
third doors	±8.0	97, 97
fourth doors	±4.1	54, 50
center door	0	34
steps by the stage	±11.0	3, 2
side stairs	±11.0	148, 146
tops of the main aisles	±4.5	117, 114

Optimal placement. Table 5 compares the optimized placement with no ushers and with 10 random placements of 6 ushers, on 100 fresh scenarios not used in the search. Local search found no improving moves, so the optimized placement is the greedy one. Two ushers stand on the side walls about 10 m to 12 m from the stage, beside the side stairs, one on each side, and direct patrons from the overloaded outermost doors to the second doors in. The other four stand along the front of the hall on the left. With the optimized ushers, casualties fall by about a third, from 1.14% to 0.79% of patrons, and survivors leave 21 steps (about 7.6 s, or 12%) sooner on average; the paired improvement in $\mathcal{J}$ is 0.0157 ± 0.0006. The ushers rebalance the doors: the outermost doors' loads fall from 229 and 237 patrons to 168 and 149, and the second doors' rise from 111 and 115 to 184 and 205, so the busiest door carries 205 patrons instead of 237.

Placement matters more than numbers. Randomly placed ushers are, on average, no better than none: their mean $\mathcal{J}$, 0.1150, is indistinguishable from the 0.1160 with no ushers. The best random placement came close to the optimum, but several were clearly harmful; the worst raised casualties to 1.41% and the survivors' mean exit time to 206 steps, worse than having no ushers at all, by sending patrons toward doors that were already overloaded. Meeting the legally required number of Crowd Managers is therefore not enough on its own: where they stand, and which doors they direct people to, determines whether they help.

Table 5: Ushers ($\bar{w} = 0.5$), 100 fresh paired replicates. $\Delta\mathcal{J}$ is the paired change relative to no ushers.

Configuration	$\mathcal{J}$	casualties	exit time	$\Delta\mathcal{J}$
no ushers	0.1160	1.14%	182.0	–
optimized	0.1003	0.79%	160.9	-0.0157 ± 0.0006
random, mean of 10	0.1150	1.05%	181.8	-0.0010
random, best	0.1017	0.81%	163.0	-0.0143 ± 0.0006
random, worst	0.1328	1.41%	206.4	$+0.0167 \pm 0.0007$

6.1.4 Combinatorial Optimization

Table 6 shows the value of each additional usher, adding them in the order chosen by the greedy search and evaluating on the same 100 fresh scenarios. The first two ushers, the symmetric pair on the side walls, provide the entire benefit: each reduces $\mathcal{J}$ by about 0.007–0.008, and together they cut casualties from 1.14% to 0.76% and survivors' mean exit time from 182 to 161 steps. The remaining four, along the front of the hall, change $\mathcal{J}$ by less than two paired standard errors each; the greedy search added them because, among 80 options evaluated on only 12 scenarios, some option always looks best by chance. Two well-placed Crowd Managers therefore capture essentially all of the benefit of redirecting patrons, well below the five that New York requires for 1010 patrons [4]. The required number is set by occupancy rather than by the needs of redirection, and the remaining managers' value lies in other duties, for example responding to an obstructed exit.

Table 6: Value of each additional usher, in greedy order ($\bar{w} = 0.5$, 100 fresh paired replicates). $\Delta\mathcal{J}$ is the paired change from one fewer usher.

Ushers	$\mathcal{J}$	casualties	exit time	$\Delta\mathcal{J}$
0	0.1160	1.14%	182.0	–
1	0.1075	0.91%	171.2	-0.0085 ± 0.0006
2	0.1003	0.76%	161.3	-0.0072 ± 0.0005
3	0.1007	0.76%	162.0	$+0.0003 \pm 0.0003$
4	0.1002	0.76%	161.0	-0.0005 ± 0.0003
5	0.1003	0.79%	160.8	$+0.0001 \pm 0.0003$
6	0.1003	0.79%	160.9	0.0000

6.1.5 Egress Times

Table 7 answers the first two questions. After a concert, with every patron leaving at once, the first floor empties in 475 time steps (171 s, just under 3 min); with a pre-movement delay of median 30 s, as patrons gather their belongings, it takes 546 steps (197 s). Most patrons leave far sooner, with a median exit time under a minute, but the final minutes are a queue: the busiest door serves about 249 patrons at 0.6 per step, which takes at least 415 steps on its own. Total egress time is therefore set by the load on the busiest door rather than by walking distance.

In a fire without Crowd Managers, the typical exit time is unchanged, but the last patrons are delayed by blocked routes, and the hall takes about 608 steps (219 s) to empty, with large variation between fires. With the optimized Crowd Managers, the busiest door's load falls from 248 to 207 patrons. Comparing the two scenarios on the same 50 fires, the Crowd Managers reduce the mean exit time by 19.7 ± 1.0 steps (7.1 s), the time by which 95% of patrons are out by 68 ± 4 steps (24 s), and casualties by 0.35 ± 0.10 percentage points. Their effect on the time the last patron leaves, -36 ± 22 steps, is not statistically significant: that time is driven by the few patrons the fire delays most, whose routes the Crowd Managers do not reach.

Table 7: Egress times for 1010 patrons, 50 replicates with common random numbers. Times are means over replicates of each replicate's statistic, in seconds; "last out" is the time the hall is empty.

Scenario	mean	median	95th pct.	last out	casualties	busiest door
concert	63.8	55.9	151.4	170.9 ± 0.7	0%	249
concert, 30 s delay	82.7	73.6	169.9	196.5 ± 2.7	0%	251
fire, no Crowd Managers	63.9	55.1	153.9	218.9 ± 8.4	1.42%	248
fire, Crowd Managers	56.8	51.8	129.5	205.7 ± 9.8	1.07%	207

These times run from the seats to the mezzanine doors; the corridors and exterior doors beyond them are not modeled here (Assumption 13). The 30 s pre-movement delay is an assumed value.

7 Sensitivity Analysis

7.1 FFCA

Fire speed and lethality. With a faster fire, engulfing the hall in a median of 3 min, and casualties occurring on first contact, casualties are far higher: 19.6% without fire avoidance and 16.5% at the optimum. The objective is again minimized at an intermediate fire weight, but the optimum shifts to $m = 2$, with $m = 3$ within two paired standard errors: a faster, deadlier fire rewards keeping farther from it, with a step away from the fire weighted about as much as a step toward an exit. The qualitative conclusion, that moderate avoidance beats both ignoring the fire and fleeing it at all costs, is robust; the optimal strength of avoidance depends on the threat.

Geometry. Moving the access points from placeholder to measured positions changed the optimal mean exit time by 16% but the optimal weights by less than 0.2, so the behavioral conclusions are robust to the geometry while the egress times are not.

8 Strengths and Weaknesses

8.1 Strengths

- Behavioral weights are learned from the model itself rather than assumed, with uncertainty quantified by bootstrap and paired validation.
- Patrons are heterogeneous in a principled way, near-optimal in a precise sense.
- All comparisons use common random numbers, so differences between scenarios reflect the scenarios rather than random variation.
- The geometry is reconstructed from the seating chart, with the access points measured.

8.2 Weaknesses

- A diagonal step takes one time step, so patrons moving diagonally walk at about 1.7 m/s and diagonal routes are somewhat optimistic.
- Only the first floor is modeled, and egress ends at the mezzanine doors rather than the street.
- All patrons are able-bodied and move at the same speed.
- Patrons start moving immediately, with no pre-movement delay, so egress times are lower bounds.
- The fire is a stylized two-dimensional threat without smoke, heat or a hot gas layer, and its spread rules come from a game rather than combustion physics.
- Fires start among the audience at random, a worst case; real assembly fires more often start on the stage or in service spaces.
- The door positions are placeholders, and the friction and dynamic field rates were chosen rather than fitted to data.

9 Conclusion

References

[1] *Fire*. Minecraft Wiki. Accessed November 8, 2026. URL: https://minecraft.fandom.com/wiki/Fire.

[2] Morgan J. Hurley et al., eds. *SFPE Handbook of Fire Protection Engineering*. 5th. Springer, 2016.

[3] Armin Seyfried et al. "Enhanced Empirical Data for the Fundamental Diagram and the Flow Through Bottlenecks". In: *Pedestrian and Evacuation Dynamics 2008*. Springer, 2010, pp. 145–156.

[4] New York State Division of Homeland Security and Emergency Services. *Who or What is a Crowd Manager?* https://www.dhses.ny.gov/crowd-manager-training. Accessed November 8, 2026.

A Code

All of the code we used for this project can be found at github.com/sidcmenon/cornell-mcm-26

We used Anthropic's Claude extensively throughout this project: to write and debug the simulation code.